Course Introduction

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Course Organization

Econ520 covers questions such as:

- Does **government spending** crowd out private investment?
  - How about government deficits?
- How does **monetary policy** work?
  - Why hasn't it worked so well lately?
  - Should we worry about inflation?
- Why does the U.S. have a **trade deficit**?
  - What could be done about it?

A key goal:

- Develop a framework for thinking about these questions coherently.
- Develop **intuition** for how the macro economy works.

Course Material

Course Website

- slides, course schedule, old exams.

Canvas site

- assignments, grades, announcements

Syllabus

- rules, grading info

Course Organization

Grading:

- 2 midterms
- 1 final exam
- several assignments (talk about those later)
- exams are written to be finished in about 2/3 of the available time

Office hours:

- Times may change (see Canvas)
- If those don't work, make an appointment (email me)

Course Contents

What we study

- Does **government spending** crowd out private investment?
  - How about government deficits?
- How does **monetary policy** work?
  - Why hasn't it worked so well lately?
  - Should we worry about inflation?
- Why does the U.S. have a **trade deficit**?
  - What could be done about it?

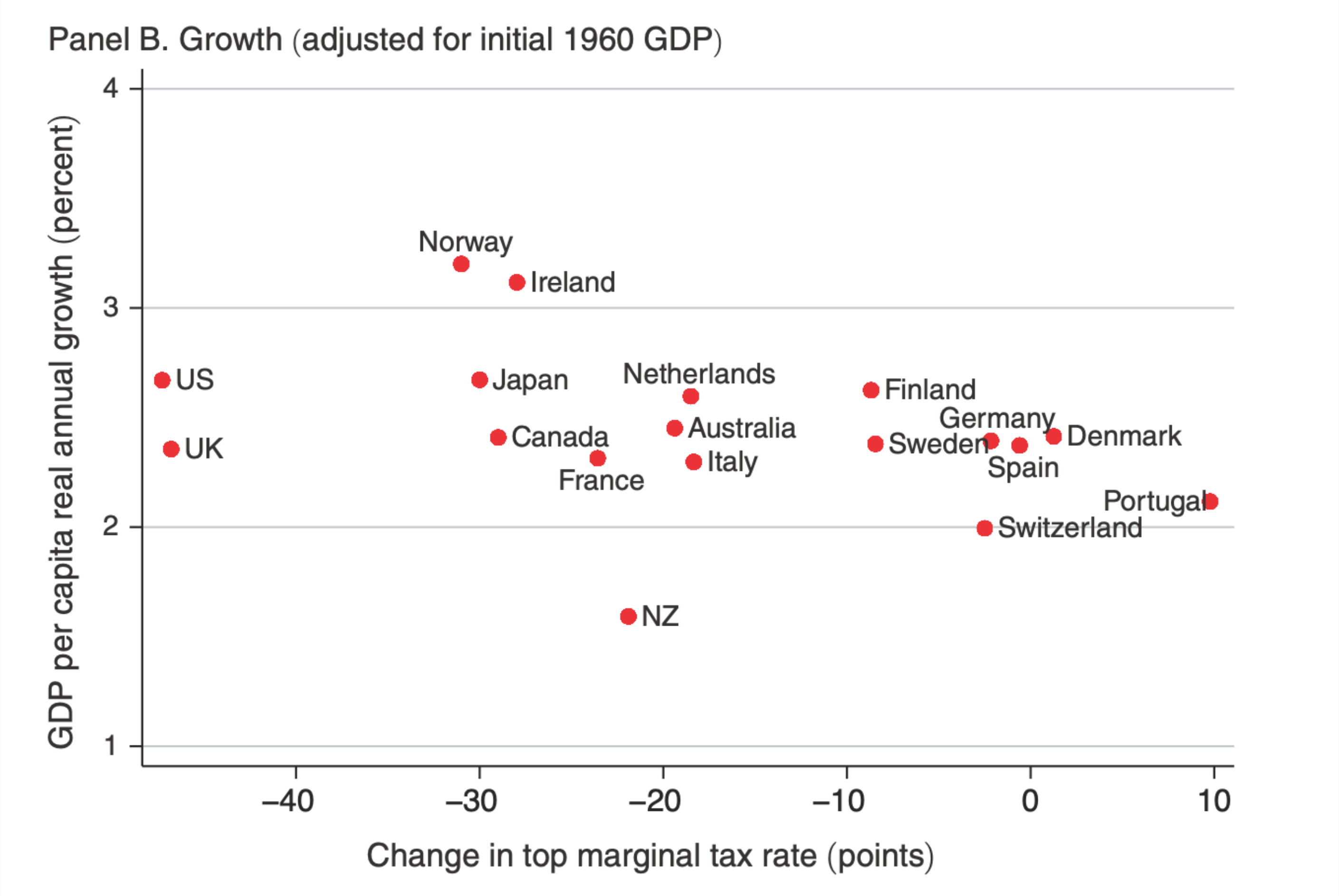
How Can We Answer such questions?

How we study these questions

Example: Do higher income taxes reduce growth?

What **methods** could we use to answer this question?

Do taxes reduce growth?



Just "look at the data." Is that convincing?

What else could we do?

Source: Piketty (2014)

Use Ordinary Least Squares (OLS) to regress
a country's growth rate on

- tax rates
- other "control variables" (education, institutions, ...)

$$\text{growth}_c = \alpha + \beta * \text{tax rate}_c + \gamma * \text{controls}_c + \varepsilon_c$$

What goes wrong?

Experiments

This is what "hard sciences" would do:

- Divide the world into **treatment** and **control** countries.
- Randomly assign each country a tax rate.
- Wait 50 years.
- Compare growth rates between high and low tax countries.

Randomized controlled trial (RCT)

- the gold standard for establishing cause-effect
- required for drug approvals

RCTs are not feasible for macro questions. – So what can we do?

Limitations of Data Analysis

> No empirical method can answer cause-effect questions without *theory*.

Think about what your stats package can see…

Models

This is why economists use models:

- We can perform thought experiments (ceteris paribus)
- "What are the effects of government debt holding everything else constant?"

Models help us to keep track of complex cause-effect chains

- Government spending $\Longrightarrow$ interest rates $\Longrightarrow$ private investment ...

Limitations of Using Models

The answer is only as good as the model.

Models are simplifications

- what to include / what to abstract from?

How to choose between competing models?

These are issues that we will discuss.

Why So Many Macro Models?

You have probably seen

- growth models (Solow, Romer, …)
- short-run IS/LM models
- medium-run AS/AD models

Why isn't there one model?

Why Isn't There One Model?

Think of these models as **special cases** of one complicated **super-model**

Each special case focuses on one set of questions:

- **short run** models:
  - prices are fixed (IS/LM)
  - business cycle (short duration) events
  - we don't worry about inflation
- **medium run** models:
  - prices adjust, but slowly (AS/AD)
  - business cycle events
- **long run** models:
  - prices are fully flexible (Solow / Romer)
  - we are interested in long-run growth

In the **short to medium run**: price adjustment **frictions**

- even nominal shocks change relative prices
- frictions give rise to unemployment, business cycles, ...
- monetary policy has **real effects**

In the **long run**: prices fully adjust

- nominal shocks only change the price level
- money becomes “neutral”
- monetary policy only affects prices; no the real economy
- aggregate demand becomes less and less important

The **AS/AD model** that we study later spells out the details.

But for now, we start simple and focus on the short run only.

The Short Run and the Long Run

Now we see why macro analysis is divided into:

- **long-run** topics
  - economic growth
  - cross-country income differences
  - focus on **supply side**
    (productivity, capital accumulation)
- short- / **medium-run** topics
  - business cycles
  - inflation and unemployment
  - focus on the **demand side**

Summary

Why do macroeconomists use models?

- Regressions don't work (omitted variables; reverse causality).
- Experiments are rarely feasible.
- Models are the fallback method for answering cause-effect questions.